

LIGHT POSTERIOR EXCHANGE FOR COMMUNICATION-EFFICIENT DISTRIBUTED LMB FUSION

Anonymous ICASSP Submission

ABSTRACT

Distributed labeled multi-Bernoulli (LMB) fusion often exchanges full Gaussian-mixture posteriors even when the receiver-side fusion step ultimately consumes label-wise moment-matched Bernoulli statistics. This paper studies a simpler communication lever: keep the effective Kullback-Leibler average (KLA) fusion graph intact, but replace full posterior messages with light posteriors containing per-label existence probabilities and single-Gaussian moments. On a 4+4 multisensor formation benchmark, a periodic light-posterior exchange matches the full-posterior baseline in tracking and consensus metrics over 50 held-out trials while reducing estimated communication by 58.6%. The result suggests that, for single-round GA/KLA-LMB fusion, payload compression can be safer than event-triggered graph sparsification.

Index Terms— Distributed multi-object tracking, labeled random finite sets, LMB filter, KLA fusion, communication-efficient sensing

1. INTRODUCTION

Distributed multi-object tracking lets a sensor network combine complementary fields of view without routing all measurements to a central processor. Labeled random finite-set filters, and in particular the labeled multi-Bernoulli (LMB) family, are attractive in this setting because they maintain object existence, kinematic uncertainty, and track identity in a single posterior representation [1, 2, 3]. The practical difficulty is communication. A multi-sensor LMB posterior can contain many labels and multiple Gaussian components per label, so repeatedly broadcasting full Gaussian-mixture posteriors is expensive for bandwidth-limited platforms.

A common response is to communicate less often or with fewer neighbors. This is appealing, but it can attack the wrong bottleneck. Conservative fusion rules such as covariance intersection, generalized covariance intersection, and Kullback-Leibler average (KLA) fusion are designed to combine partially dependent estimates without requiring unknown cross-correlations [4, 5]. In a distributed tracker, however, the quality of the fused posterior depends not only on whether the nominal graph is

connected, but also on which posterior messages actually enter the fusion step at each time. If event triggers or topology adaptation remove too many effective message paths, consensus and label agreement can degrade even while byte counts improve.

This paper takes a different position: preserve the effective fusion graph first, then reduce the payload carried by each message. The key observation is implementation-driven. In many practical GA/KLA-LMB fusion pipelines, incoming LMB components are aligned by label and reduced to label-wise Bernoulli moments before fusion. If the downstream fusion rule consumes these moments, transmitting every Gaussian-mixture component may be unnecessary in regimes where a single Gaussian moment is an adequate summary. This suggests a light posterior message containing the existence probability, mean, covariance, and minimal label metadata for each active Bernoulli component.

We study this idea in a distributed 4+4 multisensor formation benchmark. The baseline periodically exchanges full LMB posteriors over the static communication graph. The proposed light-posterior exchange keeps the same graph and synchronization schedule, but replaces each full Gaussian-mixture label density with its moment-matched single-Gaussian approximation. This design deliberately avoids graph sparsification: every baseline information path remains available to the KLA update, while communication is reduced by changing the representation of each label.

The empirical result is simple but instructive. Over 50 held-out trials, periodic light-posterior exchange reduces estimated communication by 58.6% relative to periodic full-posterior exchange, while matching the baseline local E-OSPA, consensus OSPA, position-disagreement, cardinality-dispersion, and effective-weight graph connectivity metrics. In contrast, dynamic-topology variants that alter the communication graph are less stable in the same benchmark. These results support a narrow claim: for single-round distributed GA/KLA-LMB fusion with label-wise moment matching, full posterior exchange can be overkill, and graph-preserving payload compression is a safer first lever than sparse event-triggered communication.

The contributions are:

- a graph-preserving light posterior exchange protocol for distributed LMB fusion, using label-wise moment-matched Bernoulli messages instead of full Gaussian-mixture posterior messages;
- an effective-fusion-graph perspective that separates payload compression from message-path sparsification; and
- a held-out 50-trial validation showing full-posterior-equivalent tracking and consensus with about 58% lower estimated communication in a 4+4 multisensor benchmark.

2. RELATED WORK

Labeled random finite-set tracking. Random finite-set filters provide a principled Bayesian framework for multi-object tracking with uncertain cardinality and measurement association. The probability hypothesis density filter introduced a tractable first-moment recursion for multi-object tracking [6], while labeled RFS models attach persistent identities to objects [1]. The LMB filter gives a computationally attractive labeled approximation with independent Bernoulli components [2], and multi-sensor GLMB/LMB variants extend labeled RFS tracking to distributed and cooperative sensing settings [3]. Our work does not change the LMB recursion itself; it studies how much posterior detail must be communicated before a distributed KLA-style fusion step.

Conservative distributed fusion. When local posteriors are statistically dependent and cross-correlations are unknown, conservative fusion rules are used to avoid overconfident estimates. Covariance intersection is a classical solution for Gaussian estimates [4], and KLA or generalized covariance intersection extends this log-opinion-pool principle to probability densities [5, 7]. Several RFS fusion methods apply KLA/GCI ideas to multi-Bernoulli or LMB densities, including exponential-mixture PHD fusion [8], multi-Bernoulli GCI fusion [9], and robust distributed labeled RFS fusion [10, 11]. These methods primarily define how received posteriors should be fused. We focus on the complementary question of what posterior representation should be transmitted when the fusion implementation already performs label-wise moment reduction.

Communication-aware RFS fusion. Communication constraints have been addressed through selective transmission, adaptive scheduling, consensus design, and alternative density representations. Communication-constrained decentralized fusion has been studied for labeled RFS trackers [12], consensus LMB filters reduce reliance on a central node [13], and recent adaptive fusion work adjusts weights or reliability scores to account for heterogeneous sensing and communication conditions

[14, 15]. Other work studies resilient fusion under unreliable communication or approximate average-density fusion [16]. These lines often reduce communication by changing when nodes communicate, which neighbors they use, or how much trust a received posterior receives. Our experiments show why those graph-level choices can be fragile for single-round KLA fusion: preserving the effective message graph may matter more than reducing the number of message paths.

Payload compression versus graph sparsification. The closest conceptual distinction in this paper is between transmitting fewer posterior paths and transmitting lighter posterior summaries along the same paths. Event-triggered or topology-adaptive communication can save bytes, but it also changes the effective fusion graph seen by the KLA update. A light posterior message instead leaves the graph unchanged and compresses each label density to the statistics consumed by the receiver-side fusion path. This makes the communication saving easier to attribute: any tracking or consensus change is due to posterior summarization, not to missing neighbors. The held-out results indicate that, in the tested GA/KLA-LMB setting, this payload-first design is sufficient to recover full-posterior behavior with substantially lower communication.

3. LIGHT POSTERIOR EXCHANGE

3.1. Distributed LMB Fusion Baseline

At time k , sensor i maintains an LMB posterior

$$\pi_i = \{(r_i^\ell, p_i^\ell(x)) : \ell \in \mathcal{L}_i\}, \quad (1)$$

where r_i^ℓ is the existence probability of label ℓ and $p_i^\ell(x)$ is represented by a Gaussian mixture. In the full-posterior baseline, each node periodically sends its complete GM-LMB posterior to every neighbor in a fixed communication graph. After packet drops are realized, each receiver aligns common labels and applies a single synchronous KLA-style fusion step with the same fusion weights for all compared variants.

For a receiver i and the delivered neighbor set $\mathcal{N}_i(k)$, the fusion step can be written abstractly as

$$\bar{\pi}_i \propto \pi_i^{\omega_{ii}} \prod_{j \in \mathcal{N}_i(k)} \pi_j^{\omega_{ij}}, \quad \sum_{j \in \mathcal{N}_i(k) \cup \{i\}} \omega_{ij} = 1. \quad (2)$$

The implementation performs label-wise alignment and moment matching inside the LMB fusion path. This is the reason full Gaussian-mixture transmission can be redundant: the receiver does not preserve arbitrary mixture structure after the single-round fusion update.

3.2. Light Posterior Message

The proposed message keeps the Bernoulli label set and existence probabilities but replaces each transmitted mixture density with a single moment-matched Gaussian. For a label density

$$p_i^\ell(x) = \sum_{m=1}^{M_i^\ell} w_{im}^\ell \mathcal{N}(x; m_{im}^\ell, P_{im}^\ell), \quad (3)$$

the light posterior sends

$$\tilde{p}_i^\ell(x) = \mathcal{N}(x; \bar{m}_i^\ell, \bar{P}_i^\ell), \quad (4)$$

with

$$\bar{m}_i^\ell = \sum_m w_{im}^\ell m_{im}^\ell, \quad (5)$$

$$\bar{P}_i^\ell = \sum_m w_{im}^\ell (P_{im}^\ell + (m_{im}^\ell - \bar{m}_i^\ell)(m_{im}^\ell - \bar{m}_i^\ell)^\top). \quad (6)$$

Thus a light message contains the label, existence probability, single mean, single covariance, and minimal book-keeping metadata. It does not change the local prediction, measurement update, label pruning, fusion weights, packet drop model, or neighbor schedule. Only the transmitted posterior representation changes.

3.3. Graph Preservation and Diagnostics

The communication policy is intentionally conservative. The main method sends light posteriors periodically over the same static topology used by the full-posterior baseline. This isolates payload compression from graph sparsification: if performance changes, the cause is posterior summarization, not missing message paths.

To verify this separation, we report effective-fusion-graph diagnostics in addition to tracking metrics. The attempted graph records scheduled neighbor messages, the delivered graph records successful transmissions, and the effective-weight graph records nonzero neighbor mass that actually enters the KLA update. We summarize each graph by the algebraic connectivity of its symmetrized adjacency or weight matrix. We also track label-level stale age and the total fusion weight assigned to self, light, and full posterior messages. The target behavior for a safe compression method is therefore not only low bytes, but also an effective-weight graph close to the full-posterior baseline.

4. EXPERIMENTS

4.1. Benchmark and Metrics

We evaluate on an eight-sensor 4+4 formation benchmark with 100 simulation steps per trial. Each node runs

Table 1. Held-out 50-trial results over seeds 32–81. Lower bytes and OSPA are better.

Arm	Bytes (M)	Red.	Local	Cons.	λ_2^{eff}	Pass
Full-static	27.77	0.0%	2.065	1.950	0.373	ref.
Full-dynamic	29.47	-6.1%	2.151	2.167	0.301	0/50
Light-static	11.50	58.6%	2.065	1.950	0.373	50/50
Light-dynamic	11.57	58.3%	2.067	1.964	0.371	35/50

the same local LMB prediction and measurement update, then performs one synchronous distributed fusion round. The static communication graph has 16 undirected edges. Packet delivery is stochastic, with average node-level drop probabilities near 0.2 in the held-out run. All arms use the same seeds, measurements, target trajectories, and packet-drop draws.

We report estimated communication bytes, local tracking accuracy, and network agreement. Local accuracy is measured by the mean local E-OSPA score, while network agreement is measured by consensus OSPA, position disagreement, and cardinality dispersion [17]. Communication is estimated from the number of transmitted scalar posterior values. The trial-level pass gate requires at least 30% byte reduction relative to full posterior exchange, no more than 5% local E-OSPA degradation, and no more than 10% degradation in consensus OSPA, position disagreement, and cardinality dispersion.

4.2. Compared Arms

The primary baseline is periodic full-posterior exchange on the static graph. The proposed arm is periodic light-posterior exchange on the same graph. We also include two diagnostic arms. First, full-posterior exchange with guarded dynamic topology tests whether changing neighbors alone improves the trade-off. Second, light-posterior exchange with guarded dynamic topology tests whether the light payload remains stable when the graph is allowed to adapt. The guarded dynamic topology uses the same 16-edge budget as the static graph and falls back when connectivity checks fail; it is included as an ablation rather than as the main method.

4.3. Held-out Results

Table 1 shows that the graph-preserving light posterior is the only non-reference arm that exactly matches the full-posterior baseline on both local tracking and consensus metrics. It reduces estimated communication from 27.77M to 11.50M bytes, a 58.6% reduction, while preserving the effective-weight algebraic connectivity at 0.373. Its paired trial pass count is 50/50, and its worst-case byte reduction remains above 55%.

The dynamic-topology arms clarify why the main method keeps the static graph. Dynamic topology with full posterior exchange sends slightly more bytes than the static baseline and degrades consensus OSPA by 11.1% on average; its effective-weight connectivity drops from 0.373 to 0.301. This indicates that nominally adaptive connectivity does not guarantee a better KLA information flow. Adding light payloads to guarded dynamic topology recovers most average performance, but the pass count falls to 35/50 because some trials still incur larger agreement or cardinality deviations. Thus the robust result is not “dynamic topology plus light messages,” but the simpler combination of a fixed effective graph with light posterior payloads.

These results support the paper’s central claim. In this single-round GA/KLA-LMB setting, the dominant communication saving comes from removing unneeded mixture-detail payload, not from dropping message paths. When the receiver ultimately fuses label-wise Bernoulli moments, periodic light posterior exchange can reproduce full posterior behavior while substantially reducing communication.

5. CONCLUSION

This paper studied a narrow but practical communication question for distributed LMB fusion: whether full Gaussian-mixture posteriors must be exchanged when the receiver-side KLA update uses label-wise moment-matched Bernoulli statistics. In the validated 4+4 benchmark, preserving the static effective fusion graph and sending light posterior messages matches full posterior exchange over 50 held-out trials while reducing bytes by 58.6%. The result suggests a conservative design rule for this class of trackers: preserve effective information paths first, then compress posterior payloads. Future work should identify regimes where multimodal label densities, stronger association ambiguity, or multi-round consensus require richer messages again.

6. REFERENCES

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